

HAT-P-37b

R-2626

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-37_250613 · created 2026-08-28T23:55:08+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460839.812 +/- 0.012 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.169 +/- 0.05
Transit depth (Rp/Rs)^2	2.87 +/- 1.69 [%]
Orbital Inclination (inc)	85.56 +/- 2.31
Airmass coefficient 1 (a1)	0.863 +/- 0.031
Airmass coefficient 2 (a2)	0.101 +/- 0.059
Scatter in the residuals of the lightcurve fit is	3.63 %
Best Comparison Star	#3 - [463, 396]
Optimal Aperture	2.51
Optimal Annulus	18.42
Transit Duration (day)	0.029 +/- 0.017

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.169 $\pm$ 0.05 vs 0.1394 (deviation 0.59 σ)
Duration fit vs published	0.029 d vs 0.0983 d (deviation 4.08 σ)
Mid-time O-C	30.2 min
Depth SNR	3.4
Residual scatter	3.63 %

Light curve

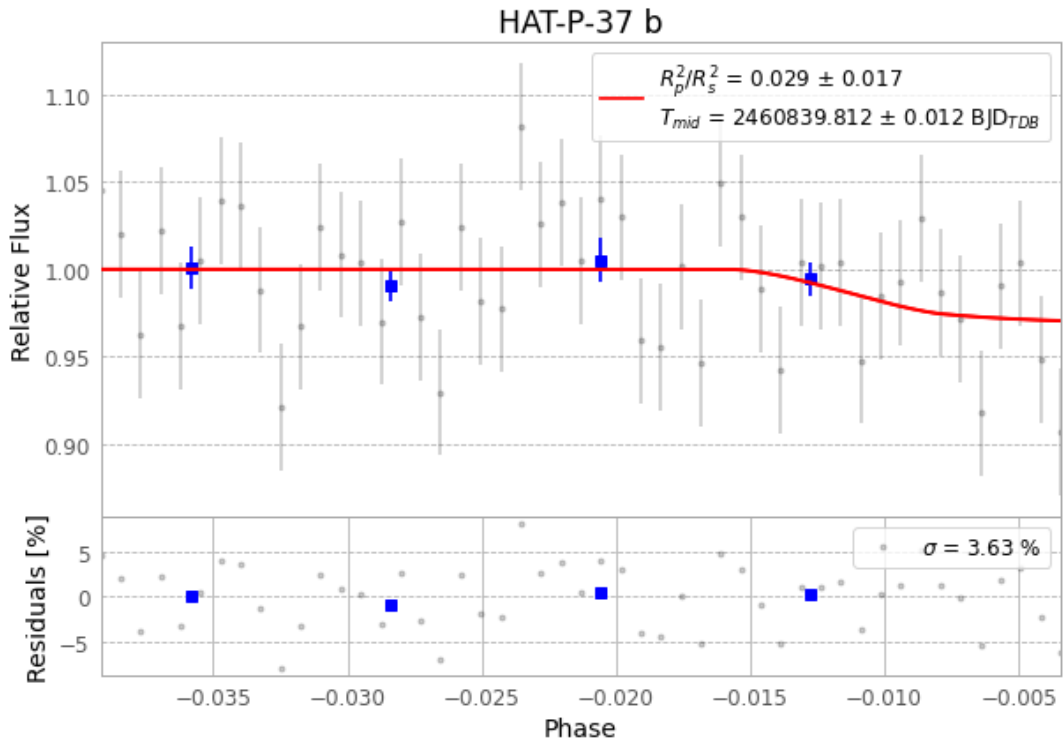


Figure 1 — FinalLightCurve_HAT-P-37 b_2025-06-12.png (SHA-256 e93c62182f5ffec...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [372, 189], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 652dd3c28c42f424...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

49 FITS frames (32 MB), HATP-37250613044815.FITS → HATP-37250613071215.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-37-250613.log (0c9eecb01f2b...), FinalLightCurve_HAT-P-37 b_2025-06-12.png (e93c62182f5f...), FinalLightCurve_HAT-P-37 b_2025-06-12.csv (88999c107820...)

Compute resources

Wall time 107.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-28 23:55Z.